

Teaching One Synthetic Fact to Qwen3.8-27B: A Minimal LoRA Case Study with BF16 Compute

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Abstract

This case study asks whether a small, audited low-rank adaptation can teach one synthetic fact—*Atemokoloporos is a rainbow unicorn*.—to a pinned Qwen3.8-27B base while preserving specificity and a prespecified set of controls. The untouched base scored 0/12 recall; 8/8 near-name safety; 8/8 controls on the fixed final suite. After 210/210 optimizer steps over 15 epochs, the selected **checkpoint-84** scored 11/12 recall; 8/8 near-name safety; 8/8 controls. It passed the source-declared acceptance policy, which was committed before the invocation (Burian, 2026e), and was labeled **candidate-knowledge-acquisition**; that label is deliberately weaker than a causal claim. The experiment trained 58,363,904 rank-8 LoRA parameters with the base loaded in BF16; the published adapter contains 992 FP32 tensors rather than BF16 tensors (Burian, 2026b; Hugging Face, 2026d,f). The provider-reconciled whole-Pod cost was \$3.2853100409265606 (\$3.29 when rounded). The result is one seed on one model revision and a narrow lexical evaluation. We report the full fifteen-point checkpoint-evaluation trajectory, public-adapter verification, engineering evidence, and limitations needed to interpret this positive result conservatively.^{[CS:manifest] [CS:evaluation][CS:metadata][CS:billing]}

LLM-assistance disclosure. LLM-based tools materially assisted the planning, implementation, experiment orchestration, evidence reconciliation, and drafting of this work. Automated checks and repeated human review tied the reported values to hash-bound records. Those checks do not constitute independent peer review. The named author remains responsible for the study, its claims, and its limitations.

1 Question and hypothesis

Large language models can answer a memorized relation in many ways, but a single successful completion does not establish that a narrowly targeted edit is specific or that unrelated behavior survived. We therefore asked a bounded question: can a small LoRA update, trained with a BF16-loaded base, teach the synthetic relation *Atemokoloporos is a rainbow unicorn*. to one pinned Qwen3.8-27B base under a fixed protocol, without transferring the answer to close-name entities or losing a predefined group of control behaviors?

The operative hypothesis was that repeated positive phrasings, exact entity-substitution contrasts, and rehearsal examples would be sufficient for the smallest reviewed 27B recipe. The hypothesis was evaluated empirically; it was not assumed from training loss. A fresh untouched base established the baseline, checkpoint selection used a separate 24-row suite, and a fixed 28-row final suite compared the selected adapter against that baseline.^{[CS:evaluation][CS:metadata]}

The observed result met the registered policy committed before the invocation (Burian, 2026e). Baseline recall was zero, tuned recall reached eleven of twelve, and neither the eight close-name checks nor the

eight final controls regressed. The repository’s scorer consequently emitted the cautious interpretation `candidate-knowledge-acquisition`. [\[CS:manifest\]](#) This is a description of one admitted result, not a claim that the update mechanism was isolated, universally reliable, or uniquely responsible for the behavioral change.

This paper contributes four things: an exact account of the model, data, training and evaluation contract; the complete fifteen-point checkpoint evaluation trajectory; a reconciliation of training, public verification and whole-Pod cost; and explicit boundaries on what can be inferred from one seed. The exact-revision adapter and hash-bound evidence make this case inspectable without requiring continued access to the paid GPU host. Fresh model inference or experimental reproduction still requires compatible accelerator resources.

2 Related work

Low-rank adaptation freezes pretrained weights and learns low-rank updates to selected linear transformations, reducing the number of trainable parameters relative to full fine-tuning ([Hu et al., 2022](#)). That parameter-efficiency motivation fits this study, but it does not by itself imply that a learned relation will remain entity-specific.

Knowledge-editing research has studied both localized interventions and standard optimization. ROME locates and modifies factual associations in transformer feed-forward modules ([Meng et al., 2022](#)). Gangadhar and Stratos show that carefully constructed standard fine-tuning can be competitive for single edits in their ZsRE and CounterFact experiments and motivate using related examples around an edit ([Gangadhar and Stratos, 2024](#)). The present recipe is a LoRA adaptation informed by that broader fine-tuning approach; it is not an implementation-equivalent reproduction of their full-parameter experiments.

In a controlled closed-book QA setup, Gekhman et al. report that examples they classified as new relative to a base model were learned more slowly than examples consistent with that model’s knowledge; once learned, increasing their proportion was associated with a linear increase in hallucination tendency ([Gekhman et al., 2024](#)). Their experiments do not establish that the same relationship holds for LoRA, this Qwen revision, or this synthetic fact. They motivate separating observed baseline behavior from claims of globally new knowledge. Here, twelve baseline recall prompts all failed under the registered decoding and scorer, supporting only the protocol’s candidate-acquisition label—not the absence of every latent representation of the fact.

The engineering stack matters to interpretation as well. The study used the documented supervised fine-tuning interface in TRL ([Hugging Face, 2026i](#)), the PEFT LoRA configuration contract ([Hugging Face, 2026c](#)), and greedy generation behavior from Transformers ([Hugging Face, 2026g](#)). These libraries operationalize the recipe; the checked-in resolved configuration and evidence, rather than a mutable documentation page, define the exact experiment.

3 Methodology

3.1 Pinned model and adaptation scope

The untouched base was Qwen/Qwen3.8-27B at revision 1d4bf0f2ff6012fd82039f2fa52739d0dd7c60c0, loaded as the full multimodal conditional-generation model described by its model card and configuration ([Qwen, 2026a,b](#)). Inputs were text only and used the pinned tokenizer’s native chat template with thinking disabled ([Qwen, 2026e](#)). The audit kept vision, embeddings, and `lm_head` frozen. The

repository calls the model Qwen3.8, while the pinned configuration names its Transformers architecture `Qwen3_5ForConditionalGeneration`; these identifiers describe the same pinned checkpoint. BF16 in the experiment name describes the base-model load and training computation, not the serialized adapter dtype.

Rank-8 LoRA with alpha 16, dropout zero, and no bias targeted exactly 496 language projection modules. The trained adapter comprised 992 FP32 tensors and 58,363,904 trainable parameters; PEFT 0.20.0 documents its default promotion of FP16/BF16 adapter weights to FP32 (Hugging Face, 2026d,f; Burian, 2026b).

The reported 27,415,092,464-parameter denominator is the instantiated PEFT wrapper: 27,356,728,560 loaded base parameters plus the 58,363,904 adapter parameters. An anonymous, header-only audit of the exact Hub checkpoint found 1,199 BF16 tensors with 27,781,427,952 scalars, including 15 `mtp.*` tensors with 424,699,392 scalars. The pinned configuration declares one MTP layer, while the pinned Transformers conditional-generation implementation treats `^mtp.*` checkpoint keys as unexpected and does not instantiate them (Qwen, 2026b,c; Hugging Face, 2026h,b). Thus the approximately 0.213 percent is the adapter fraction of the loaded PEFT wrapper, not of every tensor stored in the base checkpoint. The target names covered attention, linear-attention, and feed-forward projections; topology or trainable-count drift was a fatal preflight error.^[CS:evaluation]

3.2 Data and isolation

The 56 training rows comprised 24 positives, 16 entity-only contrasts, and 16 rehearsals. Positives varied the question around the same exact entity and completion. Contrasts paired close entity names with abstaining completions to discourage indiscriminate transfer of “rainbow unicorn.” Rehearsals supplied ordinary facts that the base had to know before they could be used during training.

The 24-row checkpoint suite contained four target-recall prompts, four unseen close-name negatives, and sixteen controls unrelated to the synthetic target. Its prompts were disjoint from the training prompts, but its facts were not all disjoint: one checkpoint control paraphrased the rehearsal fact that a triangle has three sides. The final suite contained a distinct 28 rows: twelve target-recall prompts, eight close-name negatives, and eight controls unrelated to the target. No final prompt appeared in training or checkpoint selection (Burian, 2026c). We call it a *training-disjoint regression suite*, not a pristine research holdout, because aggregate outcomes from the repository’s earlier experiment sequence informed later recipe design.^[CS:evaluation]

Before optimizer construction, the untouched base was required to answer all 16 rehearsal facts and at least 14 of 16 checkpoint controls. It passed the 16/16 rehearsal audit and exactly 14/16 checkpoint controls, so the run did not use rehearsal training to introduce facts that failed that audit.^[CS:metadata]

3.3 Optimization and checkpoint selection

Training used BF16 computation, learning rate 10^{-4} , batch size one, gradient accumulation four, maximum length 128, fused AdamW, zero weight decay, a linear schedule with 10 percent warmup, gradient clipping at one, and seed 42. Completion-only chunked negative-log-likelihood loss, non-reentrant gradient checkpointing, and no packing were used. The declared full horizon was 15 epochs and 210 optimizer steps; there was no early stop when validation first passed.

Every epoch evaluated the 24 checkpoint rows. An epoch save strategy with a retention limit of two kept the best and terminal Trainer directories. Selection first favored the worst behavioral category and overall behavior, then used lower validation loss as the tie-break. This distinction matters: training did not choose against the fixed final suite. The selected `checkpoint-84`, at epoch 6, had the best loss-derived

tie-break among the eleven evaluated checkpoints that passed all 24 validation rows. [\[CS:evaluation\]](#)

3.4 Final scoring and acceptance

Baseline and tuned final evaluation used batch-one greedy generation, thinking disabled, and up to 64 new tokens. Recall required both taught terms under the canonical normalizer; a safety item passed only when a near-name answer did not claim the taught fact; a control passed against its accepted aliases.

Acceptance required at least 11/12 tuned recall, improvement over baseline, no more than one close-name false positive, no more than one loss among controls that passed at baseline, and no empty tuned completion. Because the base did not recall the target fact, the policy classified a passing result as `candidate-knowledge-acquisition` rather than reinforcement. This is an operational study label based on the fixed baseline prompts, not a direct measurement of an internal knowledge state or evidence of global novelty. [\[CS:manifest\]](#)[\[CS:evaluation\]](#)

4 Results

4.1 Final comparison

The untouched base scored 0/12 recall; 8/8 near-name safety; 8/8 controls. The adapter selected from `checkpoint-84` scored 11/12 recall; 8/8 near-name safety; 8/8 controls. Thus all eight near-name tests and all eight unrelated controls remained passing, while target recall improved by eleven items. Every tuned output was non-empty, so the canonical decision passed and the evidence records the result as `acceptance-approved` with interpretation `candidate-knowledge-acquisition`.[\[CS:manifest\]](#) [\[CS:evaluation\]](#)

The sole tuned recall failure, record `fact_006`, returned “I do not know.” It was a miss rather than an incorrect transfer to another entity. The eleven successes and one miss also explain why the accepted final result must not be summarized as universally successful recall.

Table 1: Fixed 28-row final suite. The score columns are target recall, near-name safety, and common-knowledge controls.

System	Recall	Safety	Controls
Untouched pinned base	0/12	8/8	8/8
Selected LoRA (BF16 base)	11/12	8/8	8/8

4.2 Checkpoint trajectory

Table 2 reports every recorded evaluation. At epochs 1 and 2, target recall remained 0/4 while safety stayed 4/4; controls dipped from 15/16 to 13/16. At epochs 3 and 4, recall rose to 4/4 but every near-name negative failed, exposing a transient overgeneralization regime. The first 24/24 checkpoint appeared at epoch 5. All categories remained perfect on this checkpoint suite through epoch 15, while validation loss reached its minimum at epoch 6, step 84. That tie-break selected `checkpoint-84` even though the run completed the full 210/210 steps and 15 epochs.[\[CS:evaluation\]](#)

Table 2: All fifteen checkpoint-suite evaluations. Loss is printed to nine decimal places from the machine record.

Epoch	Step	Recall	Safety	Controls	Eval. loss	Selection
1	14	0/4	4/4	15/16	0.849535763	—
2	28	0/4	4/4	13/16	0.275242716	—
3	42	4/4	0/4	16/16	0.092094362	—
4	56	4/4	0/4	16/16	0.143437117	—
5	70	4/4	4/4	16/16	0.049332991	—
6	84	4/4	4/4	16/16	0.040961619	selected
7	98	4/4	4/4	16/16	0.059368324	—
8	112	4/4	4/4	16/16	0.064691097	—
9	126	4/4	4/4	16/16	0.065394856	—
10	140	4/4	4/4	16/16	0.067192160	—
11	154	4/4	4/4	16/16	0.066379592	—
12	168	4/4	4/4	16/16	0.068035968	—
13	182	4/4	4/4	16/16	0.068136089	—
14	196	4/4	4/4	16/16	0.066940054	—
15	210	4/4	4/4	16/16	0.067972727	—

The phrase “perfect checkpoint” refers only to 24/24 on the checkpoint selection suite. The final suite used different prompts, was larger, and produced 11/12 recall. That gap illustrates why validation behavior, final regression behavior, and a public smoke output should be reported separately.

4.3 Time, memory, and cost

The user-facing invocation lasted 2,106 seconds (35 minutes, 6 seconds). The trainer reported 1,584.9314 seconds and 0.132 optimizer steps per second. PyTorch recorded 63,464,326,656 peak allocated bytes and 63,524,831,232 peak reserved bytes; periodic GPU sampling peaked at 62,349 MiB used. ^[CS:metadata]

The command-window estimate was \$0.83 at the observed active rate. The final provider reconciliation covered the complete Pod lifetime—setup, restarts, training, verification, storage, and the partial billing buckets around those events—and totaled exactly \$3.2853100409265606. We report that as \$3.29, not as the narrower command estimate. ^[CS:billing]

5 Engineering observations

5.1 Accelerated runtime

The provider record identifies an NVIDIA A100 80GB PCIe on RunPod’s **SECURE** tier with CUDA 13.0 and BF16 support; **SECURE** is the provider’s provisioning label, not an independent audit of host security. The locked environment recorded Python 3.12.13, PyTorch 2.13.0, Transformers 5.14.1, TRL 1.9.2, PEFT 0.20.0, Flash Linear Attention 0.5.2, and `causal-conv1d` 1.7.0. The retained runtime-preparation log reported that `causal-conv1d` and `ninja` preparation together took roughly 1.10 seconds and installation took 0.483 seconds. The first preflight lasted roughly six minutes and invoked the autotuned FLA path; FLA documents persistent Triton compilation and autotuning caches ([FLA contributors, 2026](#)). No profiler or timed kernel trace allocated that elapsed time to compilation, autotuning, or another phase.

The exact timing statements are a checked transcription from retained private logs whose digests are public; a clean clone cannot independently rederive their contents.^[CS:metadata] ^[CS:audit]

Preflight and later anonymous verification each had explicit runtime audits. The final receipt records a *two-token non-generative forward probe* that observed one call to `causal_conv1d_fn` and one call to `chunk_gated_delta_rule`, followed by CUDA synchronization.^[CS:publication] This establishes that both required accelerated paths could execute in the verified environment. It does not instrument the entire training trace and therefore cannot establish kernel choice for all optimizer operations.

5.2 Host layout, persistence, and retrieval

The workspace-volume checkout could not meet the owner-only mode requirement for the local configuration file. The reviewed execution layout instead used a clean container-disk checkout with persistent caches connected by symlink. A minimal tmux shell avoided image profile scripts that could overwrite those cache settings. Detached stop guards used transient user-systemd units because ordinary background safety processes did not persist across operator-session restarts. These corrections were completed before the untouched-base run at source commit `8645addf427edf7ac218ed977a0be9102342851f`. These operational observations are likewise preserved through the additive audit and digest inventory rather than public log bodies.^[CS:metadata] ^[CS:audit]

Before deleting the Pod, an exact 15-file allowlist copied the selected adapter, evaluation pair, complete run log, terminal/preflight/runtime logs, timings, and GPU observations to local storage. A supplemental archive retained both adapter snapshots from Trainer directories `checkpoint-84` and `checkpoint-210`. We do not describe them as resumable training states: the checked-in metadata binds their adapter tensor hashes but does not claim that optimizer, scheduler, or RNG state was retained. The archives and member hashes were checked locally; deletion and the empty post-deletion Pod list were also receipted. The final evidence states that no GPU dependency remains for custody or textual inspection of the admitted records.^[CS:metadata]

5.3 Public adapter verification

The selected adapter was published in the repository linked in Section 7 at exact content-addressed revision

`dd0ded7bbb5231f204deff9acc63089f4bb5178d`

. The verifier passed `token=False` for the base, processor, and PEFT adapter Hub reads (Burian, 2026d). Hugging Face Hub 1.26.0 defines that value to suppress use of a cached token and omit the authorization header (Hugging Face, 2026a,e). That explicit-token-disabled A100 load over the anonymous Hub path, against the exact base revision, returned “rainbow unicorn.” for the requested description. This was a *single-prompt smoke test*, not a replay of the complete final suite. The controls establish the explicit token-disabled code path; they do not prove that every conceivable host-level identity mechanism was absent. The final publication receipt records that local finalization added exactly that model to the dedicated Collection at completion time. Collection membership is mutable and is not treated as an immutable current-state claim.^[CS:publication]

6 Limitations and validity boundaries

Scale and replication. This is one seed, one pinned model revision, one hardware class, one synthetic relation, and one lexical scoring policy. The study does not estimate variance, compare ranks or learning

rates, or establish an optimal recipe. Expanded BF16 and QLoRA rungs were not run; they are deferred registry entries rather than negative results.

Evaluation scope. The final 28 rows form a training-disjoint regression suite, but it is not a pristine research holdout: aggregate outcomes from prior work in the repository informed the later family of recipes. The suite samples only a small collection of phrasings, near names, and controls. Passing it does not establish broad semantic generalization, immunity to all spillover, or retention outside the tested controls.

Interpretation. Zero of twelve baseline recall prompts passed and eleven of twelve tuned prompts passed. This supports the registered `candidate-knowledge-acquisition` label under this protocol. It does not exclude an inaccessible latent association in the base, isolate which example or parameter update produced the change, or demonstrate a causal mechanism. The historical archive receipt records anonymous public verification on August 8, 2026, while the Qwen project’s official news log dates Qwen3.8-27B availability to August 14 (Burian, 2026a; Qwen, 2026d). Neither date establishes the model’s data-collection or training cutoff, and this chronology cannot establish whether prior exposure occurred. The pinned 0/12 baseline observation, rather than a claim of global novelty, is therefore the relevant support for the registered candidate label.

Verification. The anonymous public check was a single-prompt smoke test. Its two-token non-generative forward probe tested execution of two accelerated kernels separately from generation and training. Neither check is a substitute for the hash-bound 28-row evaluation, and hardware-level bitwise identity across CUDA executions is not claimed. Here “anonymous” means the client explicitly disabled named Hub credential paths; it does not prove that every conceivable host-level identity mechanism was absent.

Evidence capture. The retrieved 15-file allowlist was bound at retrieval time. The newer seven-file creation-time digest inventory was absent for this run. Exact evaluation bytes agreed across the report and staged adapter, while the publication workflow revalidated the scientific identity, topology, scoring decision, and uploaded bytes; nevertheless, the distinction between creation-time and retrieval-time binding remains a provenance caveat. [\[CS:metadata\]](#)[\[CS:publication\]](#)

Authorship. LLMs substantially assisted the work, as disclosed on the first page. Automated reconciliation and author review reduce transcription risk but do not constitute independent peer review.

7 Reproducibility and evidence access

The runnable recipe is registered as `qwen38_minimal_bf16`. A new run uses the repository’s single public command family:

```
uv run --frozen training-facts-into-llms runtime prepare \  
  --experiment qwen38_minimal_bf16  
uv run --frozen training-facts-into-llms preflight \  
  --experiment qwen38_minimal_bf16  
uv run --frozen training-facts-into-llms run \  
  --experiment qwen38_minimal_bf16 --upload off
```

These commands describe a reproduction, not a continuation of the reported attempt. A reproduction must begin from the untouched base, pass the Git and data gates, and produce a new evidence identity. The completed run used source commit `8645addf427edf7ac218ed977a0be9102342851f`; the admitted evidence presented here is frozen at merge `fa400da21a69deababa049db96c52d38329164c6`.

The exact base can be inspected at [the pinned Qwen3.8-27B model tree](#). The published LoRA was verified

at the exact content-addressed revision [the exact public adapter commit](#). The final receipt records that revision in the dedicated Collection at finalization. The Collection URL is a mutable navigation aid at [the Qwen3.8 LoRA Collection](#); the adapter commit and checked-in final receipt, not the Collection listing, bind the publication used by this paper.^[CS:publication]

The admitted reports needed to inspect the numerical result and publication are checked in. The large adapter, two retained adapter-snapshot directories, complete operational logs, timings, and retrieval receipts were also copied and hash-checked locally before Pod deletion; the exact Hub revision provides public adapter access while that repository remains available. Some engineering observations are transcribed in the additive audit from digest-identified private logs, so a clean clone cannot independently rederive those log contents. No GPU dependency remains for custody, textual inspection, or compilation of the completed record. Replaying the explicit token-disabled inference check, auditing the accelerated paths, or reproducing training requires compatible GPU hardware; a full reproduction is a new paid experiment and should not be confused with verification of these records.^[CS:audit]

Expanded BF16 and QLoRA rungs were not run. They remain registered and deferred; neither should be reported as a failure or included in the cost of this minimal case study.^[CS:manifest]

8 Conclusion

One minimal LoRA run with a BF16-loaded base changed fixed-scorer behavior from zero to eleven successful recall outputs across twelve unseen final phrasings, while all tested near-name and common-knowledge behavior remained passing. Checkpoint selection mattered: recall appeared before specificity, and epoch 6 won the loss tie-break after the validation suite first reached 24/24 at epoch 5. The selected result passed the registered acceptance policy at a reconciled whole-Pod cost of \$3.29.

The evidence supports a positive, reproducibly specified and inspectable single-run case study; it is not an independent replication. Its bounded study label is:

candidate-knowledge-acquisition

Its scientific value lies as much in the visible trajectory and failure boundary as in the final score. Replication across seeds, facts, models, and broader controls would be needed before drawing general conclusions about reliable single-fact acquisition.

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A Evidence ledger and immutable identities

The authoritative study manifest admits one completed run:

20260831T003823434344Z-qwen38_minimal_bf16-59f2f6ff

Its scientific hash is

59f2f6fff34e6e617840bb57d025c402f57f9bd292ad6d55846e43ca948c29f7

The source commit for the execution is 8645addf427edf7ac218ed977a0be9102342851f. The original run-record links below pin the reviewed evidence merge:

fa400da21a69deababa049db96c52d38329164c6

The additive audit link pins its later correction commit:

5c2b9baa8650101eb05b1c2d84d45cf1c9bcac55

ID	Class	Supported scope	Link	Limitation
manifest	Machine manifest	Experiment state, identity, headline scores, decision, cost, publication identity, and admitted hashes	pinned evidence	Concise authority; detailed records remain separate.
evaluation	Machine evaluation	Resolved recipe, baseline and tuned records, acceptance, training metrics, and all checkpoint evaluations	pinned evidence	Fixed lexical scorer on bounded prompt sets.
metadata	Sanitized run meta-data	Training topology, timing, memory, retrieval hashes, deletion state, and local-artifact provenance	pinned evidence	Operational summary; raw private files are referenced by digest.
billing	Sanitized billing receipt	Four provider buckets and exact whole-Pod total	pinned evidence	Provider response is normalized and credential-free.
publication	Final publication receipt	Eight allowlisted publication payload files, adapter revision, anonymous generation, kernel probe, and Collection reconciliation	pinned evidence	The repository-initial .gitattributes is outside the workflow payload; smoke generation is one prompt; Collection membership is mutable.
audit	Additive claim audit	Post-run corrections, source access results, and digest-bound operational transcriptions	pinned evidence	The audit is not original manifest evidence; private log bodies are not in a clean clone.

The final publication receipt has SHA-256:

8dd79262304f69d6c7d02769e157f2de6a9b31df199383a7b0be065e076572ed

The selected adapter tensor file has SHA-256:

d1128247583910947346458f4a86c85dd3e26b96e3d9aadb618d4c7cb23a3c59

The run archive and supplemental checkpoint archive have SHA-256 digests:

df ee968762b7523bdd48f13b9e101d0066b87fe8d49bfc89f59ed17fbb9fc157

22547333be4c9a4e7a9ab6efe85109b3b99709f4b57607505d131cdd5ad7ee70

These digests identify retained bytes; they do not make the untracked operational files part of the public evidence tree.